JJJ-P-630C
April 16, 1979
SUPERSEDING
Fed. Spec. JJJ-P-630B
June 6, 1969

FEDERAL SPECIFICATION

POTATOES, WHITE, DEHYDRATED

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration, for use by all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements of dehydrated white potatoes for use by all Federal Agencies and the Armed Forces as an item of general issue and as a component of operational rations.

1.2 Classification. The product shall be of the following types and styles, as specified (see 6.1):

Type I - Diced

Style A - Uncooked 1/2 x 1/2 x 1/8 inch (13 x 13 x 3 mm) quarter dice or 1/4 x 1/4 x 1/8 inch (6 x 6 x 3 mm) through 1/2 x 1/2 x 1/8 inch (13 x 13 x 3 mm) pieces of a single nominal size (Civil agencies only see 6.1)
Style B - Flakes

Type II - Mashed-precooked; rapid rehydrating

Style A - Granules
Style B - Flakes

Type III - Slices, uncooked (round, or rectangular), as specified, (see 6.1)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

FSC 8915

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Federal Specifications:

TT-C-495	- Coatings, Exterior for Tinned Food Cans
PPP-B-636	- Boxes, Shipping, Fiberboard
PPP-C-29	- Canned Subsistence Items, Packaging and Packing of
PPP-C-96	- Cans, Metal, 28 Gage and Lighter

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston; New York; Philadelphia; Washington, DC; Atlanta; Chicago; Kansas City, MO; Fort Worth; Houston; San Francisco; Los Angeles; and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-L-1497	- Labeling of Metal Cans for Subsistence Items
MIL-L-35078	- Loads, Unit; Preparation of Non-Perishable Subsistence

Military Standards:

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-668	- Sanitary Standards for Food Plants.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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Laws and Regulations

US Department of Health, Education and Welfare

Federal Food, Drug, and Cosmetic Act and Regulations Promulgation Thereunder

Visual Aids for Inspection of Metal Containters

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

US Department of Agriculture

United States Standards for Condition of Food Containers

(Application for copies should be addressed to the Information Division, Agricultural Marketing Service, US Department of Agriculture, Washington, DC 20250.)

AIC-58 New Peroxidase Test Procedure for Dehydrated Potatoes to Indicate Adequacy of Blanching

(Copies may be obtained from the Western Utilization Research and Development Division, Agricultural Research Service, United States Department of Agriculture, 800 Buchanan Street, Albany, GA 94710.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Association of Official Analytical Chemists

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, Box 540 Benjamin Franklin Station, Washington, DC 20044.)

Association of Vitamin Chemists

Methods of Vitamin Assay of the Association of Vitamin Chemists

(Application for copies should be addressed to the Interscience Publishers, a Division of John Wiley and Sons, Inc., 605 Third Avenue, New York, NY 10016.)

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National Research Council

Food Chemicals Codex (F.C.C.)

(Application for copies should be addressed to the Office of Director, National Research Council, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

American Chemical Society

Analytical Reagents

(Application for copies should be addressed to the American Chemical Society, 1155 Sixteenth St., N.W., Washington, DC 20002.)

American Dry Milk Institute, Inc.

Standards for Grades of Dry Milks, Including Methods of Analysis Bulletin 916

(Application for copies should be addressed to the American Dry Milk Institute, 130 North Franklin Street, Chicago, IL 60606.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Potatoes. The potatoes used for dehydration shall be sound, mature, of similar drying characteristics, and of mealy texture when cooked. Potatoes which discolor or become soggy after boiling shall not be used. (Potato varieties such as Russet Burbank, Green Mountain, Cobbler, Red McClure, Katahdin, Sebago, Kennebec, Pontiac, or Snowflake are recommended.)

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NOTE: In order to meet the reducing sugar requirement contained in 3.4, it is recommended that potatoes containing more than 3.0 percent reducing sugar (dry weight basis) not be used.

3.1.2 Sodium sulfite or sodium acid sulfite. The sulfite(s) shall meet the requirements of the Food Chemicals Codex.

3.1.3 Calcium salt (types I and III only). The calcium salt used shall meet the requirements of the Food Chemicals Codex.

3.1.4 Optional ingredients.

3.1.4.1 Emulsifiers and antioxidants. Emulsifiers or antioxidants, or both, which have been approved for use by the Federal Food, Drug, and Cosmetic Act may be added during processing of dehydrated white potatoes. Residuals shall comply with regulations of the Federal Food, Drug, and Cosmetic Act.

3.1.4.2 Sodium acid pyrophosphate. Sodium acid pyrophosphate meeting the requirements of the Food Chemicals Codex and in quantities which meet the regulations of the Federal Food, Drug, and Cosmetic Act may be added during processing.

3.1.5 Vitamins for fortification of type I, style B and type II, styles A and B.

3.1.5.1 Vitamin A. Vitamin A shall meet the requirements of the Food Chemicals Codex, except that it shall be a water dispersible solid form and shall not impart any fishy or other objectionable odor or flavor to the product when added at the required level.

3.1.5.2 Ascorbic acid. Ascorbic acid shall meet the requirements of the Food Chemicals Codex.

3.2 Preparation.

3.2.1 Washing and sorting. The potatoes shall be thoroughly washed to remove soil and foreign material, and sorted to remove diseased and seriously damaged units.

3.2.2 Peeling and trimming. The washed and sorted potatoes shall be peeled and carefully trimmed to remove discolored, damaged or other undesirable material.

NOTE: Unless surface enzymes are inactivated by high temperature loosening of of the peels, as with steam or lye above 180°F. (82°C.), the peeled potatoes should be given a dip in an antioxidant solution to prevent darkening on the trim table. Preheating of potatoes prior to peeling is suggested.

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3.2.3 Cutting. The potatoes shall be cut into the following nominal dimensions:

Type:

I - 1/2 by 1/2 by 1/8 inch (13 by 13 by 3 mm) quarter dice or 1/4 by 1/4 by 1/8 inch (6 by 6 by 3 mm) through 1/2 by 1/2 by 1/8 inch (13 by 13 by 3 mm) pieces to be of a single nominal size (size tolerance ± 0.04 inch (1 mm)

II - Pieces suitable for uniform cooking

III - Slices ranging in any one nominal size from 1 by 1 by 1/8 inch (25 x 25 x 3 mm) to 1-1/2 by 1-1/2 by 1/8 inch (38 x 38 x 3 mm), or 1/8 inch (3 mm) cross sectional cuts from potatoes with a diameter of 2-1/2 inches (64 mm) or less, as specified, (see 6.1). If the potatoes are larger than 2-1/2 inches (64 mm) in diameter, they shall be either cut in half or quartered longitudinally prior to slicing

NOTE: Free starch should be washed off after the cutting operation.

3.3 Processing.

3.3.1 Blanching (type I, style A and type III only). The type I, style A and type III potatoes shall be adequately blanched before dehydration to meet the requirements of 3.4.1.1 and 3.4.1.4, respectively.

3.3.2 Cooking (type I, style B, only). Cooking shall be by such method(s) as to result in an end product having the quality characteristics stipulated in 3.4.

NOTE: Boiling water or atmospheric steam cook 212°F (100°C), when employing bed depths of 1-inch (25 mm) or less for 18 - 20 minutes has been satisfactory.

3.3.3 Cooking (type II, styles A and B). The potatoes shall be thoroughly cooked but not overcooked prior to mashing. Cooking shall be by such method(s) as to result in an end product having the quality characteristics stipulated in 3.4.

3.3.4 Sulfiting (all types). The potatoes shall receive a sulfite treatment by such method(s) as will insure uniform distribution throughout the product, and a residual sulfur dioxide content in the dehydrated product within the limitations specified for the respective types (see 3.4.1.1 through 3.4.1.4).

3.3.5 Calcium treatment (types I and III only). When specified (see 6.1), the diced or sliced product shall be subjected to a calcium salt solution, or a combination of calcium and sodium bisulfite solution (see 3.3.4), of such concentration and for such time as to obtain a uniform distribution throughout the product, and a residual calcium content in the finished product as specified for the respective types (see 3.4.1.1 and 3.4.1.4).

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3.3.6 Mashing and mixing (type II only). The cooked potatoes shall be mashed and prepared for drying in granular or flake form, without destroying the cellular structure of the potatoes. Optional ingredients (see 3.1.4.1 and 3.1.4.2), if incorporated into the mash shall be by such method as to insure uniform distribution.

3.3.7 Dehydration. Dehydration shall be by such methods as to result in a finished product complying with 3.4. The dehydration condition shall be such that the products will not be scorched or otherwise damaged by heat.

3.4 Finished products.

3.4.1 Dehydrated products. The products except type II, style A (see 3.4.1.2) shall possess a bright, uniform white potato color which may range from a very light cream to pale yellow. The products shall be free from haylike, scorched, rancid, musty, or other objectionable odors. Unless otherwise specified (see 6.1), the moisture content shall not exceed 6.0 percent by weight, for product intended for use overseas or afloat or in standard "B" rations or authorized reserve stokage programs, nor 8.0 percent by weight for product intended for more rapid turnover and for civilian agencies. Unless otherwise specified, (see 6.1), the reducing sugar level shall not be greater than 3.0 percent (dry weight basis). The products shall be free from insect infestation or evidence thereof or of other foreign materials, such as splinters, paint, varnish, rubber or metals. When specified (see 6.1), vitamin fortification shall be accomplished prior to dehydration for type I, style B and type II, style B potatoes and either prior to or after dehydration for the type II, style A potatoes in such a manner as to insure uniform distribution, and at such levels as to result in one ounce (28 g) of the dehydrated product containing not less than:

Vitamin A	1000 International units
Ascorbic acid	50 mg.

Unless otherwise specified, the product shall be packaged not more than 90 days before shipment.

3.4.1.1 Type I. Unless otherwise specified (see 6.1) the sulfite content (calculated as sulfur dioxide) shall be 450 $\pm$ 150 p.p.m. for style A and 350 $\pm$ 150 p.p.m. for style B when specified and the calcium content for both styles shall be 0.3 $\pm$ 0.1 percent, by weight. For Military agencies: Not more than 10.0 percent, by weight, of the product shall pass through a US Standard No. 4 sieve, and not more than 1.0 percent, by weight, of the total, shall pass through a US Standard No. 8 sieve, when tested as specified in 4.5.7. One hundred grams of the product shall not contain more than 20 black or dark brown spots greater in area than a circle 1/32 inch (1 mm) in diameter or lightly discolored areas greater in area than a circle 1/4 inch (6 mm) in diameter or more than 5 black or dark brown spots greater in area than a circle 1/8 inch (3 mm) in diameter. The total of black or dark brown spots less in area than a circle 1/32 inch (1 mm) in diameter and lightly discolored spots less in area than a circle 1/4 inch (6 mm) in

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diameter shall not adversely affect the appearance of the product. Slight discoloration, such as a light grayish tinge to the majority of the units shall not be considered a defect but shall be judged under the color factor of the rehydrated and cooked product. The type I, style A product shall have a color score of 50 or less when tested in accordance with 4.5.3 for peroxidase activity. For Civil agencies: Not more than 1.0 percent by weight of the product shall pass through a US Standard No. 8 sieve, when tested as specified in 4.5.7. The product shall not contain more than 3.0 percent, by weight, of defective pieces. Any piece of product shall be considered defective if it shows any area affected by heat damage (scorching), or other factors causing discoloration, as evidenced by an orange, pink, red, brown or black color. Other defective pieces are those which show an objectionable and noticeable area of pathological injury, peel, eye, bruise, dark discoloration caused by contact with the dehydration trays or belts, or by adherence of organic material thereon, or other types of discoloration. Minute defects are not to be considered as defects unless the aggregate materially detracts from the overall appearance of the product in the dry or cooked form. Slight discoloration such as light grayish tinge to the majority of the pieces shall not be considered a defect but shall be judged under the color factor of the rehydrated and cooked product.

3.4.1.2 Type II, style A. Unless otherwise specified (see 6.1) the sulfite content (calculated as sulfur dioxide) shall be 350 ± 150 p.p.m. The product shall be in a granular form and of such particle size that 100 percent shall pass through a US Standard No. 40 sieve when tested as specified in 4.5.7. The color shall be similar to the "good" color standard which is the color desired, but shall not be darker than the "borderline" color standard (see 4.5.5). 1/ When examined in accordance with table VI, the type II, style A product shall not contain black or brown specks in excess of the following: Military agencies - 20 on the average and 30 in a unit sample or Civil agencies - 25 on the average and 35 in an individual subsample. When specified the product shall contain not less than 5 percent milk solids from nonfat dry milk, whole dried milk or a combination of both, or dry milk substitute. If a dry milk substitute is used, the minimum protein content of the substitute shall be no less than 27.0 percent and the maximum fat content shall be 2.5 percent.

3.4.1.3 Type II, style B. Unless otherwise specified (see 6.1) the sulfite content (calculated as sulfur dioxide) shall be 350 ± 150 p.p.m. Flakes shall be of such size that not more than 10.0 percent, by weight, shall pass through a US Standard No. 40 sieve, when tested as specified in 4.5.7. When examined in accordance with table VI, 100 grams of the type II, style B product shall not contain black, brown or orange specks over 1/16 inch (2 mm) in any dimension in excess of the following: Military agencies - 10 on the average and 15 in an individual subsample or Civil agencies - 15 on the average and 20 in a unit sample. Product with a vitamin C enrichment in accordance with good commercial practice is acceptable.

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3.4.1.4 Type III. Unless otherwise specified (see 6.1) the sulfite content (calculated as sulfur dioxide) shall be 200 - 600 p.p.m. and the calcium content shall be 0.3 ± 0.1 percent, by weight for product in #10 can. Not less than 75 percent, by weight, of the product shall be retained on a US Standard 1/2 inch screen (13 mm) and not more than 5.0 percent, by weight, of the product shall pass through a US Standard 1/4 inch screen (6 mm) when tested as specified in 4.5.7. One hundred grams of the product shall not contain more than 20 black or dark brown spots greater in area than a circle 1/32 inch (1 mm) in diameter or lightly discolored areas greater in area than a circle 1/4 inch (6 mm) in diameter or more than 5 black or dark brown spots greater in area than a circle 1/8 inch (3 mm) in diameter. The total of black or dark brown spots less in area than a circle 1/32 inch (1 mm) in diameter and lightly discolored spots less in area than a circle of 1/4 inch (6 mm) in diameter shall not adversely affect the appearance of the product. Slight discoloration, such as a light grayish tinge to the majority of the units, shall not be considered a defect but shall be judged under the color factor of the rehydrated and cooked product. The type III product shall have a color score of 50 or less when tested in accordance with 4.5.3 for peroxidase activity.

3.4.2 Rehydrated and cooked products.

3.4.2.1 Rehydration and cooking.

3.4.2.1.1 Type I, style A. The product shall rehydrate and cook to approximately its original shape and size after cutting (see 3.2.3), when 4 ounces (113 g) of the dehydrated product are added to 24 fluid ounces (710 ml) of cold water in a cooking vessel, covered, brought quickly to a boil, and then simmered for not more than 25 minutes.

1/ Color standards may be obtained from the US Army Natick Research and Development Command, Food Engineering Laboratory, ATTN: DRDNA-W, Natick, MA 01760.

3.4.2.1.2 Type I, style B. The product shall rehydrate to approximately its original shape and size after cutting (see 3.2.3) when 24 fluid ounces (710 ml) of boiling water are added to 4 ounces (113 g) of dry product, stirred, covered and allowed to stand for 15 minutes.

3.4.2.1.3 Type II, style A. The product shall rehydrate to a mashed potato consistency when 8 fluid ounces (236 ml) of hot water is added to 2 ounces (57 g) of dehydrated material and whipped with a fork or wire beater for 1 minute. Boiling water shall be removed from heat and volumetrically measured and cooled to not greater than 180°F (82°C) before adding to the potatoes.

3.4.2.1.4 Type II, style B. The product shall rehydrate to a mashed potato consistency when the dry product is added to the hot water and reconstituted with label instructions.

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3.4.2.1.5 Type III. The product shall rehydrate and cook to approximately its original shape and size after cutting (see 3.2.3), when 4 ounces (113 g) of the dehydrated product are added to 24 fluid ounces (710 ml) of cold water in a cooking vessel, covered, brought quickly to a boil, then simmered for not more than 25 minutes.

3.4.2.2 Color. The products shall possess a typical cooked white potato color and shall show no objectionable darkening within 10 minutes after preparation.

3.4.2.3 Flavor and odor. The products shall possess a normal cooked white potato flavor and shall be free from objectionable sulfite and other foreign flavors and odors.

3.4.2.4 Texture and character.

3.4.2.4.1 Type I. The product, after cooking or rehydration (see 3.4.2.1.1 and 3.4.2.1.2), shall possess a mealy texture.

3.4.2.4.2 Type II. Type II rehydrated product (see 3.4.2.1.3 and 3.4.2.1.4) shall possess a typical fluffy mashed-potato texture and shall be free from gumminess, pastiness, and lumps.

3.4.2.4.3 Type III. The product, after cooking (see 3.4.2.1.5), shall possess a mealy texture.

3.5 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

3.6 Workmanship.

3.6.1 For civil agencies. The processing plant, grounds, equipment, packaging materials, personnel practices, and sanitary practices used in production of this item shall be such as to minimize the possibility of contamination of the product through microbial growth, dust, pests, rodents, condensate, or other unsanitary sources. The product shall be prepared and processed with minimum delay between the various stages of production.

3.6.2 For military agencies. The product shall be processed in establishments meeting the requirements of MIL-STD-668. It shall be prepared, processed, and dehydrated with minimum delay between the various stages of production.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling. Samples of the finished product covered by this specification, and the samples of any materials, components, or constituents entering into the preparation of the finished product, may be taken at any time by the Government inspector to determine compliance with the requirements.

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4.2 Inspection.

4.2.1 General. Inspection shall be made by the procuring agency or a duly authorized representative, at the time and place designated by the procuring agency. It may be made at the site of manufacture, both during and after processing and packaging, at any suitable point in transit, and after delivery to the point of destination.

4.2.2 Factory inspection. The passing as satisfactory of any detail of processing or materials shall not relieve the supplier of responsibility for faulty workmanship or materials which may be discovered at any time prior to final acceptance.

4.2.3 Final inspection. Unless otherwise specified (see 6.5 and 6.6), final inspection shall be made after delivery to the point of destination.

4.2.4 Responsibility for inspection. The supplier is responsible for meeting all the provisions of the specification prior to submission of supplies to the Government for acceptance. Unless otherwise specified in the contract or order, the supplier is required to perform the examination and tests prescribed herein, using his own or any other inspection facilities and services acceptable to the Government.

4.2.4.1 Inspection records of the examination and tests shall be complete and made available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.3 Preacceptance inspection (plant sanitation inspection).

4.3.1 Civil agencies. Inspection of plants producing the product will be conducted upon the request of the contracting officer to the appropriate Government inspection agency. Award of contracts will be limited to establishments known to maintain proper sanitary conditions or which have received prior sanitary approval recommended by the inspection agency.

4.3.2 Military agencies. Product shall be rejected if produced in plants not meeting the requirements of 3.6.2.

4.4 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.4.1 Inspection of components and materials (examination and testing). In accordance with 4.2.4 and 4.2.4.1 above, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

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4.4.1.1 Testing of components. Testing of components described in this specification shall be as shown in table I. Results shall be applicable to the lot average. Nonconformance to test requirement shall be cause for rejection of the lot.

TABLE I. Testing of components 1/

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported	Test ref and rqmt para
Can 2/	4 by 4 inch (102 x 102 mm) panel and cover (cans-603 by 700 or larger)	Cans	S-1	Tinplate	Nearest 0.01 lb (gram)/ base box	5.1.1.1 & 4.5.11
	One can with lid (smaller than 603 by 700 can)					5.1.1.2.1

1/ Each component or component characteristic, as applicable, which is not listed in this table, unless otherwise excepted shall be tested in accordance with the applicable procedure contained in the pertinent subsidiary specification.

2/ Testing of can coating (when required) and can labeling shall be performed on the same cans submitted for tinplate test. One or more failures shall be cause for rejection of the lot.

4.4.2 Process examination. When specified (see 6.1), examination shall be performed during preparation and processing to determine conformance to required procedures and controls specified in 3.2 and 3.3. Nonconformance to procedures and controls shall be cause for rejection of the involved finished product.

4.4.3 Examination of end item. Examination shall be in accordance with tables III through VII and 4.4.3.1 through 4.4.3.5 (and 4.4.3.6 when applicable). The sampling plans and acceptance criteria shall be as shown in table II. Acceptable quality levels (AQLs) shall be expressed in terms of defects per hundred units for tables VI and VII and percent defective for tables III, IV and V.

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TABLE II. Sampling plans and AQLs

Table	Inspection level	Sample unit	Lot size expressed in	AQLs Major	AQLs Minor
III	S-3	1 filled primary container	Primary containers	-	2.5
IV	S-3	1 filled and sealed 5 gallon can (18.9 L)	5 gallon cans (18.9 L)	0.65	-
V	S-2	1 filled primary container	Primary containers	-	2.5
VI	S-2	8 oz. (227 g) from each primary container	Primary containers	1.0	-
VII	I	1 filled and closed package	Packages	1.5	4.0

TABLE III. Examination for net weight 1/

Category	Defect
Minor	
201	Net weight more than 5 percent under specified net weight container required to have less than 1 pound (454 g)).
202	Net weight more than 2 percent under specified net weight (container required to have 1 pound (454 g) or more).

1/ Report results to nearest 1/8 ounce (gram) for containers required to have less than 1 pound (454 g) and to nearest 1/4 ounce (gram) for containers required to have 1 pound (454 g) or more. Lot shall be rejected if sample data indicates a lot average net weight less than specified net weight.

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TABLE IV. Examination of 5 gallon (18.9 L) cans for leakage 1/

Category	Defect
Major	
151	Leakage.

1/ See 4.5.10.2 for procedure.

TABLE V. Examination for particle size (see 4.5.7)

Category	Defect
Minor	
201	Particle size of dehydrated product not conforming to requirements.

TABLE VI. Examination of finished product 1/

Category	Defect
Major	Dehydrated
151	Not bright, uniform, white potato color, ranging from very light cream to pale yellow (applicable to type I and III and type II, style B).
152	Haylike, scorched, rancid, musty, or other objectionable odor. 2/
153	Color darker than borderline color standard (applicable to type II, style A) (see 4.5.5).
154	More than 20 black or dark brown spots each greater in area than a circle 1/32 inch (1 mm) in diameter or lightly discolored areas greater in area than a circle 1/4 inch (6 mm) in diameter, in any combination, in 100 grams of product (applicable to type I or III). 2/
155	More than 5 black or dark brown spots each greater in area than a circle 1/8 inch (3 mm) in diameter in 100 grams of product (applicable to type I or III). 2/

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TABLE VI. Examination of finished product 1/ (cont'd)

Category	Defect
Major	Dehydrated (cont'd)
156	The number of black or brown specks exceeds that specified for 100 grams of product (applicable to type II, style A). 2/ 3/ 5/
157	The total of black or dark brown spots less in area than a circle 1/32 inch (1 mm) and lightly discolored spots less in area than a circle 1/4 inch (6 mm) in diameter in 100 grams of product adversely affects the appearance of the product (applicable to type I or III). 2/
158	The number of black, brown or orange specks exceeds that specified for 100 grams of product (applicable to type II, style B). 2/ 4/ 5/
	Rehydrated (see 3.4.2.1.1 through 3.4.2.1.5 for procedures as applicable)
159	Does not rehydrate to a mashed potato consistency (applicable to type II).
160	Color, flavor, odor, texture, shape (as applicable to specified type) not typical of normal cooked white potato.
161	More than a trace of peel.

1/ Presence of foreign material such as but not limited to glass, wood, metal, paint, insect, insect fragment, or other extraneous materials shall be cause for rejection of the entire lot.

2/ Make one determination per sample unit.

3/ a. Check each sample separately.
b. Fill a 100 mm diameter cell of a double Petri dish to overflowing with granules.
c. Draw a straight edge across the top of the Petri dish to form a level surface.
d. Examine entire area of top surface and count all black and brown specks regardless of size. Ignore tan or light brown specks.

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4/ a. Check each sample separately.
b. Extract 100 grams of flakes from a sample and examine on a white tray in small increments, taking care to avoid thick layers.
c. Examine entire 100 grams, while counting all the black, brown or orange specks over 1/16 inch (2 mm) in dimension.

5/ The lot shall be rejected if the average sample data indicate the number of specks exceeds the specified number. See 3.4.1.2 and 3.4.1.3.

TABLE VII. Examination of package

Category		Defect
Major	Minor	
151		Tear or hole through package or open seal.
152		Not closed.
	201	Not clean.
153		Nomenclature missing, incorrect, or illegible.
	202	Other required label information missing, incorrect, or illegible.

4.4.3.1 Examination of cans. External examination of cans and examination for leakage, except for 5 gallon cans (18.9 L) (see 4.5.10.2), shall be in accordance with the applicable examination criteria contained in PPP-C-29. Visual Aids for Inspection of Metal Containers (illustration of defects only) shall be used for classification of defects.

4.4.3.2 Examination of can labeling (military agencies). Examination of can for labeling shall be in accordance with the examination criteria contained in MIL-L-1497.

4.4.3.3 Examination of cans for labeling (Civil agencies). The can shall be examined to determine compliance with 5.4.2 as pertains to labeling. The sample unit for examination shall be one can. The sample size shall be the number of sample units indicated by inspection level S-1. The lot size shall be expressed in cans. The acceptable quality level shall be 2.5 expressed as percent defective. A can not conforming to all label requirements shall be a defective unit.

4.4.3.4 Examination of can coating. When exterior coating of can is required to be in accordance with TT-C-495, examination of can coating shall be in accordance with the examination criteria of that specification.

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4.4.3.5 Examination of shipping container for level A, B and C. Examination of the filled and closed shipping container shall be in accordance with the examination criteria in the appendix of PPP-B-636. In addition, the following defects shall be included in the table of examination: Major: Nomenclature, NSN, Contract number and date of pack, missing, incorrect or illegible. Minor: Other required marking, missing, incorrect, or illegible. When level C packing is required to be in accordance with the Uniform Freight C' ssification or National Motor Freight Classification, only the additional defects, specified above, for marking and arrangement shall apply. Cases intended for transfer at sea shall not be strapped with found wire.

4.4.3.6 Examination of unit loads. When shipping containers are required to be in unit loads in accordance with MIL-L-35078, examination of such loads shall be in accordance with that specification.

4.4.4 Sampling procedure and acceptance criteria for testing of cans for oxygen content (type II only). Procedures for testing of oxygen shall be in accordance with 4.5.9. The sample unit shall be one can. Lot size shall be expressed in terms of cans and the sample size for testing shall be the number of sample units indicated by inspection level S-2. The acceptable quality level, expressed as percent defect, shall be 2.5. A sample unit with more than 2 percent oxygen shall be scored as a defect. Test results shall be reported to the nearest 0.1 percent.

4.4.5 Sampling procedure and acceptance criteria for testing of finished product. Procedures for testing of moisture, reducing sugars (dry weight basis), calcium (types I and III), sulfite, vitamins (when required), and peroxidase activity (type I, style A, and type III), shall be in accordance with 4.5. The sample for testing shall be a 1 pound (454 g) composite derived from the number of cans indicated by inspection level S-2. Lot size shall be expressed in cans. Each result shall be reported to the same decimal place as specified in the pertinent requirements. One or more test failures shall be cause for rejection of the lot.

4.4.6 Condition inspection of containers. When inspection of the product for quality and condition is performed by the US Department of Agriculture, the United States Standards for Conditions of Food Containers shall apply. (For applicable criteria governing inspection of containers for the Military, see 4.4.3).

4.5 Tests. Test procedures and controls which differ from those specified herein may be used by the supplier if they provide a quality assurance equivalent to that specified. If the Government contracting officer determines that such procedures and controls do not provide, as a minimum, such quality assurance, the supplier will use the test procedures set forth herein. In case of dispute as to test results, the test methods specified herein shall govern.

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4.5.1 Moisture. Moisture determination shall be made by the vacuum-oven method as specified in 4.5.1.1 and 4.5.1.2. Samples for moisture determination shall be collected in clean, dry jars which shall be completely filled and sealed with airtight lids.

4.5.1.1 Preparation of sample. Prepare a clean dry Waring blender or similar mill by grinding a sample and then discarding it. Grind 1/ approximately 100 grams of the well-mixed sample in the prepared Waring blender or similar mill for one minute. Place the ground material in a small, dry, tightly closed container for at least one-half hour. Pass the ground material over a 20 mesh screen nested on a 40 mesh screen. That portion which passes through the 20 mesh screen but not the 40 mesh screen is to be analyzed for moisture, and shall be weighed immediately or placed in a small, dry container with a tightly closed lid until ready for analysis (a one-half pint Mason jar is satisfactory).

4.5.1.2 Procedure. Weigh two, approximately 2 gram, samples of the ground screened material (that portion which has passed through the 20 mesh screen and has been collected on the 40 mesh screen) into tared, dry, aluminum weighing dishes approximately 2 or 2-1/2 inches (51 or 64 mm) in diameter and 3/4 inch (19 mm) in depth with tight-fitting covers. Place the dishes with cocked lids in a vacuum-oven and dry for 6 hours at 70°C. under a pressure of not more than 100 mm of mercury. During drying, admit to the oven a slow current of air (approximately 2 bubbles per second), dried by passing through concentrated sulfuric acid. Remove the dishes and allow them to cool in a desiccator before weighing. Calculate the percent of moisture in the original samples.

4.5.2 Sulfite. Sulfite (calculated as sulfur dioxide) shall be determined on the potatoes ground as specified in 4.5.1.1 but not sieved, according to the method specified in the Official Methods of Analysis of the Association of Official Analytical Chemists, Chapter: Food Additives: Direct, section: Chemical Preservatives, method. Total Sulfurous Acid, Modified Monier-Williams.

4.5.3 Peroxidase activity. The test to show adequate blanching shall be made in accordance with the following method:

(a) Reagents. Reagents shall be Reagent Grade, Analytical Reagent or Chemically Pure as defined by the American Chemical Society in the A.C.S. Analytical Reagents or meet the requirements of the Food Chemicals Codex.

10 percent guaiacol in 95 percent undenatured ethyl alcohol.

30 percent hydrogen peroxide (keep the stock bottle of 30 percent hydrogen peroxide in a cool place and pour a small amount into a small dropping bottle for daily use).

5M ammonium acetate (386 gm of $(NH_4)C_2H_3O_2$ made up to 1 liter with distilled water.

Potassium dichromate ($K_2Cr_2O_7$).

Cobalt nitrate ($Co(NO_3)_2 \cdot 6H_2O$).

1/ Type II, style A potatoes need not be ground.

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(b) Apparatus.

Comparator block.

Set of color standards (color scores 0, 50, 100, 200 and 300) and an extra color standard of color score 50.

Six 1 liter beakers.

Twenty-four 18 by 150 mm pyrex or heat-resistant test tubes.

Three 10 ml pipettes.

Six 65 mm funnels.

Test tube rack.

15 cm fluted filter paper (Whatman No. 12 or similar grade).

Six stirring rods (glass rods or aluminum or plastic spoons are satisfactory, do not use iron).

Containers for the ammonium acetate, guaiacol, and hydrogen peroxide.

30 or 60 ml dropping bottle.

1 liter graduated cylinder.

Thermometer, 0° to 110°C.

(c) Color standards*

1. Prepare a 300 color score standard by making up 0.24 gm of $K_2CR_2O_7$, and 15.6 gm of $Co(NO_3)_2 \cdot 6H_2O$ to a liter of solution.

2. Prepare a 200 color score standard by diluting 100 ml of the 300 color score standard to 150 ml.

3. Prepare a 100 color score standard by diluting 100 ml of the 300 color score standard to 300 ml.

4. Prepare a 50 color score standard by diluting 100 ml of the 300 color score standard to 600 ml.

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5. The 0 color score standard is distilled water.

6. Fill color standards into the 18 by 150 mm test tubes and seal to prevent evaporation.

* Care should be exercised in preparing standards and making dilutions.

(d) Procedure.

1. Place a 50.0 gm sample of dehydrated potatoes in a liter beaker.

2. Add 10 ml of 5M ammonium acetate, followed by 540 ml of distilled water at 23.5 ± 3.5°C.

3. Stir thoroughly with a stirring rod and allow to rehydrate for 1 hour (control temperature at 23.5 ± 3.5°C.).

4. After the hour of rehydration, add 10 ml of 10 percent guaiacol and stir.

5. Add 6 drops of 30 percent hydrogen peroxide, stir thoroughly and allow to react for exactly 1 hour without agitation. To prevent fading of color, protect sample from sunlight and intense artificial light by covering the beakers with a dark cloth or towel during the reaction.

6. At the end of the hour of reaction time*, stir thoroughly by swirling in several directions with a stirring rod and take care to insure that the color is uniformly distributed throughout the solution.

7. Filter a portion of the sample through Whatman No. 12 fluted filter paper placed in a 65 mm funnel into an 18 by 150 mm test tube.

8. Estimate extent of coloration by comparison with color standards in a comparator block or test tube rack with a white background. Make comparisons by looking through the diameter of the tube and not the length. Samples that give a color score of 50 or less are adequately blanched.

* Stirring, filtering and comparison of color must be completed within 5 minutes due to the tendency of the color to fade.

(For further information see AIC-58, New Peroxidase Test Procedure for Dehydrated Potatoes to Indicate Adequacy of Blanching.)

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4.5.4 Calcium. Test for calcium shall be performed as follows:

Reagents shall be Reagent Grade Analytical Reagent or Chemically Pure as defined by the American Chemical Society in the A.C.S. Analytical Reagents or meet the requirements of the Food Chemicals Codex.

1. Grind sample in Wiley or similar mill (20 mesh size or smaller).
2. Place aliquot (or 25 grams) of ground sample in 250 ml capacity dish (Vycor or similar).
3. Heat over open flame until major portion of carbon is burned away.
4. Ash sample in muffle furnace (550°C.) to white or nearly white ash.
5. Cool ash.
6. Wet with small amount of distilled water.
7. Add 2 ml concentrated HCl.
8. Transfer quantitatively into 250 ml beaker using 100 ml of distilled water for washing.
9. Add 2 or 3 drops of methyl red indicator and adjust to red color with hydrochloric acid.
10. Heat solution to boiling.
11. Add 15 ml of saturated ammonium oxalate to gently boiling solution (see note).
12. While continuing a gentle boil, stir in 2 heaping tablespoons of dry urea.
13. Cover with watch glass.
14. Continue boiling until color changes from red to yellow at pH 5.0 of methyl red indicator (requires approximately 15 minutes).
15. Cool solution to room temperature and filter solution through medium porosity sintered glass filter.
16. Wash precipitate with 15 ml of 2 percent (by volume) ammonium hydroxide solution cooled to 5°C.
17. Dissolve calcium oxalate precipitate in 100 ml of hot 5 percent H_2SO_4 solution and transfer to original beaker.

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18. Heat solution to 60°C., add 2 or 3 drops of a standardized (0.01-0.1N) potassium permanganate solution, wait until pink color disappears and then complete titration with the potassium permanganate solution.

19. Titrate also a reagent blank consisting of 150 ml of hot 5 percent sulfuric acid.

20. Calculate as percent calcium.

NOTE: If solution becomes turbid, add concentrated HCl, dropwise, until clear.

4.5.5 Color (type II, style A only). A sample of the material shall be placed in a 2 ounce (60 cc) square flint glass bottle for comparison with the "good" and "borderline" color standards. Color comparisons shall be made in an illumination booth as shown in figure 1. Illumination shall be effected by two 15 watt florescent strips with a light quality of illuminant C and a level of illumination of approximately 200 foot candles. Observation shall be made with the sample and standard placed midway in the booth and inclined at a 45 degree angle from the vertical (back).

4.5.6 Reducing sugars.

Reagents shall be Reagent Grade, Analytical Reagents or Chemically Pure as defined by the American Chemical Society in the A.C.S. Analytical Reagents or meets the requirement of the Food Chemicals Codex.

(a) Preparation of sample:

1. Samples other than potato granules shall be ground in a Waring or similar type blender and sieved to retain that portion which passes through a US Standard No. 20 sieve. Raw potatoes, if tested in response to the recommendation in the note under 3.1.1, need not be sieved. Potato granules need not be ground nor sieved.

2. Weigh 25 grams of potato granules or ground sieved material into a 250 ml centrifuge bottle, add 75 ml of water, mix and allow to stand for 1/2 hour. Fifty (50) grams of the ground raw potatoes shall be used without dilution.

3. Add 100 ml of 95 percent ethyl alcohol plus 1 gram of calcium carbonate.

4. Heat in a hot water (90°C.) bath for 1 hour while stirring frequently.

5. Centrifuge sample at 1500 rpm for 10 minutes and decant the liquid portion into a 600 ml beaker.

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6. Repeat extraction three more times, each time using 75 ml of 80 percent ethyl alcohol, heating in the hot water (90°C.) bath for 15 minutes while stirring frequently, centrifuging at 1500 rpm for 10 minutes and decanting the liquid into the 600 ml beaker.

7. Evaporate combined extracts on a steam bath to a volume of 50 to 75 ml.

8. Transfer extract into 250 ml volumetric flask with hot distilled water, cool, add 10 ml of saturated, neutral lead acetate and bring to volume.

9. Filter liquid through No. 2V Whatman filter paper into 250 ml Erlenmeyer flask containing about 4 gms of potassium oxalate.

10. Mix filtrate thoroughly and allow precipitate to settle. Check filtrate for deleading by adding a few potassium oxalate crystals.

11. Decant liquid through No. 2V Whatman filter paper into 250 ml glass-stoppered Erlenmeyer flask.

(b) Procedure:

Determine reducing sugar according to the method of Munson-Walker, as described in the Official Methods of Analysis of the Association of Official Analytical Chemists; chapter: Sugar and Sugar Products, section: Invert Sugar-Chemical Methods. Calculate as percent invert sugar, dry weight basis.

4.5.7 Screening. Screening test shall be performed as follows: Weigh entire contents of primary container. Place approximately 4 or 8 avoirdupois ounces (113 or 227 g) of type I, II, style B or III product or 1 or 2 pounds (454 or 907 g) of type II, style A, product on an 8 inch (203 mm) or 12 inch (305 mm) diameter, respectively, US Standard Screen of appropriate sieve size(s) (see 3.4.1.1, 3.4.1.2, 3.4.1.3 and 3.4.1.4) and tip back and forth through a 90 degree arc 25 times. Where the use of more than 1 screen is indicated, the sieve with the larger opening should be placed on top. Empty materials remaining on sieve(s) into appropriate container(s) and repeat the procedure until the entire contents of the primary container have been screened. Weigh total material passing through sieve(s) and calculate percentage to the nearest whole percent.

4.5.8 Vitamin A and ascorbic acid. Determinations shall be made in accordance with the following methods described in Methods of Vitamin Assay: Vitamin A - Carr-Price blue color colorimetric method and ascorbic acid-2, 6 dichlorophenolindophenol photometric method with the Loeffler and Ponting modification.

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4.5.9 Oxygen content. Oxygen content of the gasses in the filled and sealed can shall be determined in accordance with American Dry Milk Institute, Inc. publication, Bulletin 916, Standards for Grades of Dry Milks, Including Methods of Analysis; Chapter: Methods of Analyses; Determination of Oxygen.

4.5.10 Leakage tests.

4.5.10.1 Size 603 by 700 and 401 by 411 cans. The can shall be tested for leakage by submerging in water contained in a desiccator, or other suitable container, and maintaining a vacuum of 10 inches (254 mm) of mercury (atmospheric pressure 29.9 inches (760 mm)), for at least 30 seconds. A leak is indicated by a steady progression of bubbles. Isolated bubbles caused by entrapped air are not considered as signs of leakage.

4.5.10.2 Five-gallon (18.9 L) cans. The filled and closed can shall be tested for leakage by submerging the can in water in such a manner that the can is lying on a side, with not less than 3 inches (76 mm) of water over any portion of the can. The can shall be held in this position for not less than 30 seconds, and shall then be rotated so that observation for leaks can be made with each of the four sides facing the top of the tank. The temperature of the water shall be not less than 30°F (17°C) higher than the air temperature of the room at time of testing, but in no case shall the temperature of the water be less than 100°F (38°C). The can shall be submerged for at least 2 minutes. A leak is indicated by a steady progression of bubbles. Isolated bubbles caused by entrapped air shall not be considered as signs of leakage.

4.5.11 Tin coating weights. Tin coating weights shall be determined by any method specified in PPP-C-96.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The product shall be packaged level A, B or C, as specified (see 6.1).

5.1.1 Level A. Unless otherwise specified (see 6.1), the product, type, style, and amount as shown in table VIII, shall be packaged in accordance with 5.1.1.1, 5.1.1.2, or 5.1.1.3, as applicable.

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TABLE VIII. Product type, style, net weight per can, number, and arrangement of cans in each shipping container

	Type	Style	Can size	Net weight Pounds	Ounces	Kg	Number in case	Arrangement Length	Width	Depth
**	I	A	603 x 700	2	4	1.02	6	3	2	1
							12	3	2	2
	I	B	603 x 700	1	10	0.74	6	3	2	1
							12	3	2	2
	I	B	5 gal.* (18.9 L)	9	4	4.20	2	2	1	1
	II	A	401 x 411	1	8	0.68	24	4	3	2
**	II	A	603 x 700	6	2	2.78	6	3	2	1
	II	A	603 x 700	6	-	2.72	6	3	2	1
	II	B	603 x 700	2	8	1.13	6	3	2	1
							12	3	2	2
**	III	-	603 x 700	1	4	0.57	6	3	2	1
							12	3	2	2
**	III	-	5 gal.* (18.9 L)	7	0	3.18	2	2	1	1

* When specified (see 6.1).

** This type and style is exclusive to Military agencies and the moisture content of this product is 6 percent or less. This commodity has been designed principally for use afloat, overseas, Standard Army "B" rations or for extended storage.

5.1.1.1 Type I product. The type I product shall be filled into a size 603 by 700 open-top style, round metal can with soldered side seams and compound-lined, double-seamed ends. The can shall be made throughout from not less than commercial 0.25 pound (113 g) per base box electrolytic tin plate (see 4.5.11), and shall be coated outside with a coating conforming to type I or II or, when specified (see 6.1), type III, of TT-C-495. The can shall be hermetically sealed. The filled and hermetically sealed can shall not show leakage when tested in accordance with 4.5.10.1. When specified (see 6.1), 9 pounds and 4 ounces (4.20 kg) of

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the style B product shall be filled into a hermetically sealed 5 gallon (18.9 L), square metal can having the dimensions of 9-3/8 by 9-3/8 by 13-7/8 (or 13-3/4) inches (238 x 238 x 352 mm (or 349 mm)) with soldered side seams and double-seamed soldered or squeezed-on soldered top and bottom. The can shall be made throughout from not less than commercial 0.25 pound (113 g) per base box electrolytic tin plate (see 4.5.11). Alternatively, the can top shall be provided with a cover not less than 6-3/16 inches (157 mm) in diameter, compound-lined and double-seamed in position, or a cover 6-7/8 inches (175 mm) in diameter, compound-lined and rolled or clinched over a flared flange opening in the can, forming a joint inclining outwards approximately 20 degrees from the vertical. The 5 gallon (18.9 L) cans shall be coated outside with a coating conforming to type I or II or when specified (see 6.1), type III of TT-C-495. The 5 gallon (18.9 L) cans shall not leak when tested in accordance with 4.5.10.2.

5.1.1.2 Type II product. Six pounds (2.72 kg), of the type II, style A product or 2 pounds, 8 ounces (1.13 kg) of type II, style B product shall be packaged in a size 603 by 700 can as specified in 5.1.1.1 and table VIII, except that prior to sealing, the product shall be gas packaged by first removing the air from the filled can and replacing it with nitrogen. If tip soldering is used to close the exhaust vents for nitrogen filling, a suitable noncorrosive flux shall be used. When the end of the can or lid is clinched prior to vacuumizing and gassing, the body flange or lid shall be indented or channeled to prevent seal-off trouble. The oxygen content of the gases at whatever pressure is in the filled and sealed container, not sooner than 2 days nor more than 14 days after filling, shall not exceed 2.0 percent when tested in accordance with 4.5.9.

5.1.1.2.1 Type II, style A product. When specified (see 6.1), twenty-four ounces (0.68 kg) of type II, style A product shall be filled into a size 401 by 411, open-top, round metal can with soldered side seam and compound-lined, double-seamed ends. The can shall be made throughout from not less than commercial 0.25 pound (113 g) per base box electrolytic tinplate (see 4.5.11), and shall be coated overall outside with a coating conforming to type I or, when specified (see 6.1), type III of TT-C-495. The product shall be gas packaged in accordance with 5.1.1.2.

5.1.1.3 Type III product. One pound, four ounces (0.57 kg) of the type III product shall be packaged in a size 603 by 700 can as specified in 5.1.1.1. When specified (see 6.1), 7 pounds (3.18 kg) of the product shall be filled into a 5 gallon (18.9 L) square can as specified in 5.1.1.1.

5.1.2 Level B (for civil agencies). The product shall be packaged in accordance with 5.1.1, except that cans with or without a commercial exterior coating will be acceptable. When specified (see 6.1), 2, 2-1/4, 2-1/2 or 5 pounds (0.91, 1.02, 1.13 or 2.27 kg) of type II or type III product shall be packaged in a sealed commercial moisture barrier Kraft or polyethylene bag.

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5.1.2.1 When specified (see 6.1), 15 ounces (425 g) of the type II, style A product shall be gas packaged in commercial packages of the type normally used by the manufacturer for this product, or not less than 6 pounds (2.72 kg) of type II, style A product shall be gas packaged in a 603 x 700 can, as specified in 5.1.1.2.

5.1.3 Packaging, level C. Five pounds (2.25 kg) of the product shall be packaged in a poly laminated bag or kraft bag in accordance with good commercial practice.

5.1.3.1 Level C (for military agencies). The product shall be packaged in cans in accordance with 5.1.1, except that the cans with or without commercial exterior coating will be acceptable and that type II, style A product shall be gas packaged in accordance with 5.1.1.2.

5.2 Packing. The canned product in the number and arrangement of cans as specified in table VIII shall be packed on end in a shipping container in accordance with level A, B, or C, as specified (see 6.1). Type I, style B product in 5 gallon (18.9 L) cans shall be packed level C only. Type III product in 5 gallon (18.9 L) cans shall be packed level B or C only. The style RSC shipping containers may be top-opening or end opening. Reinforcement of end-opening containers shall be in the same relationship to the contents as the top-opening containers.

5.2.1 Level A. The shipping containers shall be a snug-fitting fiberboard box constructed, closed and reinforced in accordance with style RSC, V2s, of PPP-B-636.

5.2.2 Level B. The shipping container shall be a snug-fitting fiberboard box constructed, closed and reinforced in accordance with style RSC, V3c, V3s or V4s of PPP-B-636.

5.2.2.1 When specified (see 6.1 and 6.2), the shipping container specified in 5.2.2 shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive, filament reinforced tape in accordance with the appendix of PPP-B-636, except that two reinforcing bands may be used, one lengthwise and one girthwise.

5.2.3 Level C. Four 5 pound (2.25 kg) bags of product shall be packaged on end within a snug-fitting fiberboard shipping container constructed and closed in accordance with style RSC, grade V3c of PPP-B-636. Each container shall be provided with full height 1/2 inch (13 mm) slotted partitions fabricated from fiberboard conforming th grade V15c of PPP-B-636. The partitions shall divide the container into four equal cells. Each shipping container shall be reinforced with nonmetallic strapping or pressure-sensitive filament reinforced tape in accordance with the appendix of PPP-B-636, except that two reinforcing bands may be used, one lengthwise and one girthwise.

5.2.3.1 When specified, (see 6.1 and 6.3), the shipping container shall be in accordance with Uniform Freight Classification or National Motor Freight Classification, as applicable.

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5.3 Unit loads. When specified (see 6.1), the product packed as specified in 5.2, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified.

5.4 Labeling and marking.

5.4.1 Military agencies.

5.4.1.1 Cans. Cans shall be labeled in accordance with MIL-L-1497. In addition, the cans shall be lithographed or printed on the lid or body of the can with the following information included in the labeling:

Nomenclature
Type_____Style_____(as applicable)
Preserved with Sulfur Dioxide
Calcium stabilized (types I and II only)
Net weight of can
Contractor's name, address (City and State)
Direction for Use (if inside the can, so indicate (see 5.4.1.1.1, 5.4.1.1.2, and 5.4.1.1.3)
GAS PACKED ITEM (applicable to type II only)

5.4.1.1.1 Directions for use, (603 by 700 can). The following directions for use, as applicable, shall be lithographed on the body or on one end of the can, or printed on the label. Alternatively, the following directions for use shall be printed on a sheet of good quality paper and one sheet placed over the product in each can after filling. The printed sheet shall not impart an odor or flavor to the product.

Potatoes, White, Dehydrated, Diced, Uncooked
(type I, style A)

Buttered Potatoes, Diced

Yield: 33 servings — Each portion: 5-1/4 ounces (3/4 cup)

Ingredients	Weights	Measures	Method
Potatoes, dehydrated, diced	2-1/4 lbs.	1 no. 10 can	1. Add potates to salted water. Cover. Bring quickly to a boil.
Water	—	7 quarts	2. Simmer until potatoes are tender, but not mushy, about 25 minutes.
Salt	2 oz.	3 tbsp.	3. Drain, if necessary.
Butter	2/3 pound	1-1/3 cups	4. Add butter and pepper and mix.
Pepper, black		1/4 tsp.	

NOTE: Substitute 1 pound (454 g) dehydrated diced potatoes for 4-2/3 pounds (2.1 kg) raw (peeled) potatoes.

Reconstituted cooked diced potatoes may be used in any recipe requiring diced potatoes, such as hashed brown, au gratin, or lyonnaise potatoes, salads, hash, stews or soups.

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Potatoes, White, Dehydrated, Diced, Cooked
(type I, style B)

Buttered Potatoes, Diced

Yield: 25 servings — Each portion: 5 ounces (about 2/3 cup)

Ingredients	Weights	Measures	Method
Potatoes, cooked, dehydrated	1 lb. 10 oz.	1 no. 10 can	1. Add potatoes and salt to boiling water. Stir.
Water, boiling	--	1-1/4 gal.	2. Remove from heat, cover and let stand 20 minutes.
Salt	1-1/3 oz.	2 tbsp.	
Butter or margarine	4 oz.	1/2 cup	3. Drain. Add butter and pepper. Stir gently.
Pepper, black	--	1/4 tsp.	

NOTE: Substitute 1 pound (454 g) precooked dehydrated diced potatoes for 4-2/3 pounds (2.1 kg) raw (peeled) potatoes.

Reconstituted cooked diced potatoes may be used in any recipe requiring diced potatoes, such as hashed brown, au gratin, or lyonnaise potatoes, salads, hash, stews or soups.

5.4.1.1.1.1 Labeling. Commercial labeling which meets the requirements of the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder shall be acceptable. Contractor's Directions for Use are required including number of servings per bag.

5.4.1.1.2 Directions for use (401 by 411 can). The following directions for use shall be lithographed on the body or one end of the can, or printed on the label:

Mashed Potatoes
(type II, style A)

Yield: 25 servings — Each portion: 5 ounces (about 2/3 cup)

Ingredients	Weights	Measures	Method
Water, warm		3 cups	1. Put warm water into bowl. Stir milk into water to reconstitute milk.
Milk, nonfat, dry	3 oz.	2/3 cup	
Water, boiling		2-1/2 qts.	2. Add water, butter or margarine, and seasonings to reconstituted milk. Stir to combine.
Butter or margarine	4 oz.	1/2 cup	
Salt		1 tbsp.	
Pepper		1/4 tsp.	
Potato granules	1-1/2 lbs.	3-1/4 cups (1 - 401 by 411 can)	3. Pour potato granules rapidly into hot liquid while stirring. Scrape down bowl.
			4. Whip until fluffy.

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NOTE 1: Alternate method: Place quantities of boiling and warm water in the mixing bowl. Mix nonfat dry milk, salt, pepper and potato granules together. Pour mixture rapidly into hot liquid while stirring. Scrape down bowl. Add butter or margarine. Whip until fluffy.

NOTE 2: Consistency may be controlled by taking out 2 cups of hot water before adding potatoes. Add the 2 cups of hot water during the whipping until proper consistency is obtained.

NOTE 3: Substitute 1 pound (454 g) dehydrated granules for 4-2/3 pounds (2.1 kg) mashed potatoes.

5.4.1.1.3 Directions for use (603 by 700 can). The following directions for use shall be lithographed on the body or one end of the can, or printed on the label: Alternatively, the following directions for use shall be printed on a sheet of good quality paper and one sheet placed over the product in each can after filling. The printed sheet shall not impart an odor or flavor to the product.

Mashed Potatoes
(type II, style A)

Yield: 100 servings

Each portion: 5 ounces (about 2/3 cup)

Ingredients	Weights	Measures	Method
Milk, nonfat, dry	12 oz.	2-3/4 cups	1. Blend potatoes and milk together, Set aside for step 3.
Potato granules	6 lb. 2 oz.	3 qts. (1 no. 10 can)	
Water boiling		13-1/2 qts.	2. Place water, butter or margarine, and salt in mixer bowl. Stir to combine.
Butter or margarine	1 lb. 2 oz.	2 cups	
Salt	2 oz.	3 tbsp.	
Pepper		2 tsp.	3. At low speed, rapidly add potato and milk mixture to liquid; mix 1/2 minute; scrape down bowl.
			4. Add pepper. Whip at high speed for about 2 minutes or until light and fluffy.

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NOTE 1: Consistency may be controlled by reserving 2 quarts (1.9 liters) hot water before adding potatoes. Add 2 quarts (1.9 liters) of hot water during whipping until proper consistency is obtained.

NOTE 2: If potatoes become dry while standing on steam table, add hot water or hot milk and mix.

NOTE 3: Substitute 1 pound (454 g) of dehydrated granules for 4-2/3 pounds (2.12 kg) mashed potatoes.

NOTE 4: For type II, style A product with specified 8 percent moisture, commercial directions for use are required, including the approximate number of servings per can.

Mashed Potatoes
(type II, style B)

Preparation directions to be supplied by the manufacturer

Potatoes White, Dehydrated, Sliced, Uncooked
(type III)

Basic Potato Recipe

Yield: 20 servings

Each portion: 5 ounces (about 3/4 cup)

Ingredients	Weights	Measures
Potatoes, dehydrated, sliced	1-1/4 lbs.	1 no. 10 can
Water	--	1 gallon
Salt	1 oz.	1-1/2 tbsp.

Method

1. Add potatoes to salted water. Cover. Bring quickly to a boil.
2. Simmer until potatoes are tender, but not mushy, about 25 minutes.
3. Drain, if necessary.

NOTE: Substitute 1 pound (454 g) dehydrated sliced potatoes for 4-2/3 pounds (2.1 kg) raw (peeled) potatoes.

Reconstituted sliced potatoes may be used in any recipe requiring sliced potatoes such as hashed brown, au gratin, or lyonnaise potatoes, or salad.

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5.4.1.2 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129.

5.4.1.3 Unit loads. Unit loads shall be marked in accordance with MIL-STD-129 and MIL-L-35078.

5.4.2 Civil agencies.

5.4.2.1 Unit containers. Commercial labeling of the unit containers shall comply with the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder. Contractor's direction for use are required including number of servings per bag.

5.4.2.2 Shipping containers. Shipping containers shall be marked in accordance with the requirements of FED-STD-123 and such other information as specified by individual agencies.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and style of product required (see 1.2).
(c) When type III is specified, whether round 1/8 inch (3 mm) cross sectional slices from potatoes with a diameter of 2-1/2 inches (64 mm) or less or 1/8 inch (3 mm) half or quarter cross sectional slices from potatoes with a diameter greater than 2-1/2 inches (64 mm) or rectangular slices ranging in any one nominal size from 1 by 1 by 1/8 inch to 1-1/2 by 1-1/2 by 1/8 inch (25 x 25 x 3 mm to 38 x 38 x 3 mm) are required (see 1.2 and 3.2.3).
(d) Military procurement of type I and II potatoes intended for use overseas or afloat or in standard "B" rations or authorized reserve stoakage programs shall be treated with calcium (see 3.3.5, 3.4.1.1 and 3.4.1.4).
(e) When vitamin fortification for type I, style B or type II, style A or B is to be added (see 3.4.1).
(f) When moisture content may exceed 6.0 percent (see 3.4.1).
(g) When nonfat dry milk solids, whole dry milk, a combination of both or dry milk substitute is required in formulation of the product (see 3.4.1.2).
(h) When sulfite content is other than specified (see 3.4.1.1, 3.4.1.2, 3.4.1.3, and 3.4.1.4).
(i) When it is required that examination be performed during prepaation and processing to determine conformance to required procedures and controls (see 4.4.2).
(j) Levels of packaging and packing required (see 5.1 and 5.2).
(k) When type III exterior can coating is required (see 5.1.1.1 and 5.1.1.2.1).

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(l) When type I or III product is to be packaged in 5 gallon (18.9 L) cans (see 5.1.1.1 and 5.1.1.3).
(m) When type II style A product is to be packaged in a size other than 401 by 411 can (see 5.1.1.2.1).
(n) When 2, 2-1/4, 2-1/2 or 5 pounds (0.91, 1.02, 1.13 or 2.27 kg) of type II or type III product is to be packaged in a sealed commercial Kraft or polyethylene bag (see 5.1.2).
(o) When other than 15 ounces (4.25 g) or 6 lbs. (2.72 kg) of type II style A product are to be packaged in commercial packages (see 5.1.2.1).
(p) When packing specified in 5.2.2.1 or 5.2.3.2 is required.
(q) Type and class of unit load when unit loading is specified (see 5.3).
(r) Quantities required (see 5.1.2 and 5.1.2.1) Civil agencies.

6.2 Packing specified in 5.2.2.1 is intended for transfer at sea operations or specific overseas operations.

6.3 Packing specified in 5.2.3.1 is intended for direct shipment from the supplier to the first consuming activity for immediate use within CONUS.

6.4 If samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.5 Suppliers inspection. Suppliers should be required to assure that the product conforms with the specification prior to submission to the Government for final inspection.

6.6 Destination inspection (For civil agencies only). When the finished product has been inspected and passed at point other than destination, the contract should require that the product be inspected at destination for condition and quantity only.

6.7 Military agencies. Based on conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700.15/NAVSUPINST 4030.28/ AFR 71-6/MCO 4030.14D/ or DSAR 4145.7, as applicable.

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MILITARY CUSTODIANS:

Army - GL
Navy - SA
Air Force - 50

Review activities:

Army - MD
Navy - MC, MS
DP - SS

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS
USDA - AMS
HEW - FDA
VA - DMS

Preparing activity:

Army - GL

Project No. 8915-0593

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein.

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